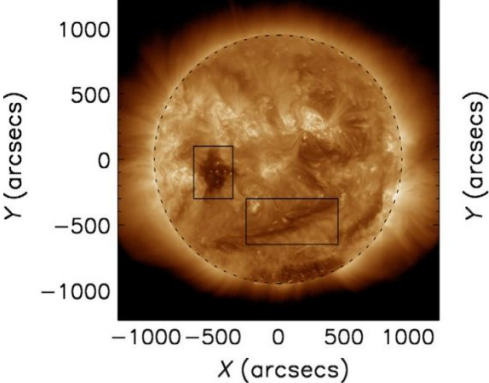


Véronique Delouille, Stefan J. Hofmeister, Martin A. Reiss, Benjamin Mampaey, Manuela Temmer, Astrid Veronig, Chapter 15 - **Coronal Holes Detection Using Supervised Classification**, Editor(s): Enrico Camporeale, Simon Wing, Jay R. Johnson, Machine Learning Techniques for Space Weather, Elsevier, 2018, Pages 365-395, ISBN 9780128117880,

Coronal holes is an important factor in studying space weather(for instance, high-speed solar wind streams), so it is important to identify coronal holes accurately. This work aims identify coronal holes accurately as well as avoid misclassification of filament.

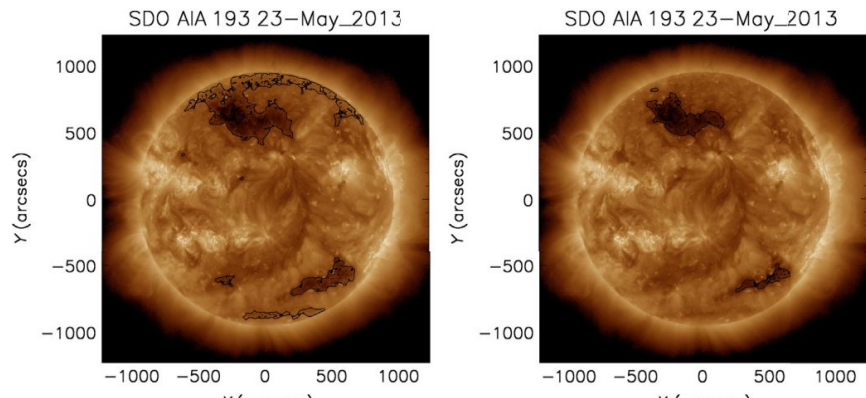
In this work, we use Solar Dynamics Observatory (SDO) Atmospheric Imaging Assembly (AIA) 193Å images (Lemen et al., 2012) and SDO/Helioseismic and Magnetic Imager (HMI) magnetograms (Scherrer et al., 2012) in order to analyze the properties of coronal holes and filament channels during the period August 19, 2010 until December 31, 2016.



AIA 193Å image taken on May 13, 2015 with a coronal hole and filament outlined (both are dark/have low intensity!)

Verbeeck, C., et al. "The SPoCA-suite: Software for extraction, characterization, and tracking of active regions and coronal holes on EUV images." *Astronomy & Astrophysics* 561 (2014): A29. is a set of segmentation procedures that allows decomposition of an EUV image into regions of similar intensity.

This algorithm is modified by taking a more conservative approach when classifying a pixel as belonging to the low-intensity class. Then manually label the low-intensity class into coronal holes and filaments..



Sets of Attributes Used for Classification

1. Set 1: Location, first- and second-order statistics on AIA images (“AIA” dataset)
2. Set 2: Location, shape measures, and first- and second-order statistics on AIA images (“AIA, shape” dataset)
3. Set 3: The “AIA” dataset augmented by the first- and second-order statistics on HMI LOS images (“AIA, HMI” dataset)
4. Set 4: The “AIA, HMI” dataset, and in addition, the shape measures (“AIA, shape, HMI” dataset)

SUPERVISED CLASSIFICATION

The extracted regions were manually labeled as either FC or CH regions by visually inspecting H α filtergrams from the Kanzelhöhe Solar Observatory for binary classification.

The proposed dataset has an imbalance ratio of 1:15

Solutions:

1: Cost-Sensitive Learning: more weight for the minority class in loss function

2: Sampling Methods: Siedhoff, D., 2015. ADASYN algorithm to reduce class imbalance by synthesizing minority class examples. Available from:

<https://nl.mathworks.com/matlabcentral/fileexchange/50541-adasyn-improves-class-balance-extensionof-s mote>

3: Ensemble Learning: use the Matlab function *fitcensemble* with decision trees as base classifiers and compare the results of the aggregating methods AdaBoost.M1 and RUSBoost.

Model training

Perform 50 iterations of the following protocol. We do a stratified shuffle-split of the original dataset into a 75% development (“training”) set and 25% evaluation (“test”) set. To choose the best hyperparameter combination, use stratified fivefold cross-validation.

Perform metrics:

True Skill Statistics

$$\frac{TP}{TP+FN} - \frac{FP}{FP+TN}$$

Receiver Operating Characteristic

Results

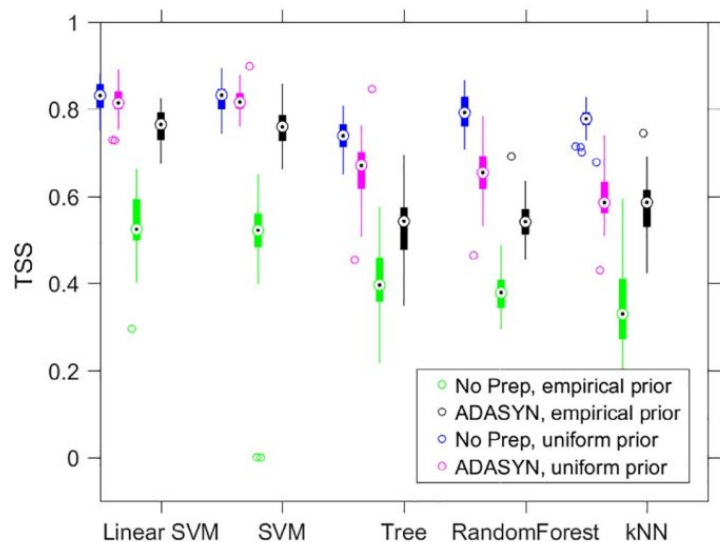
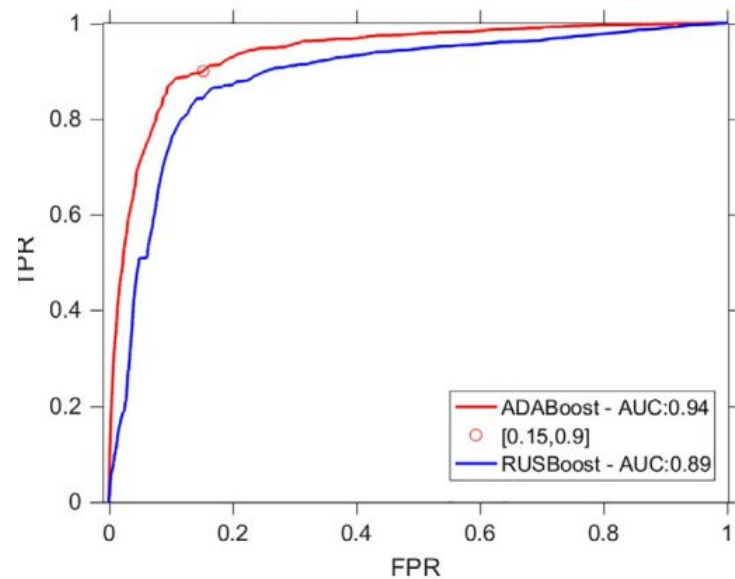
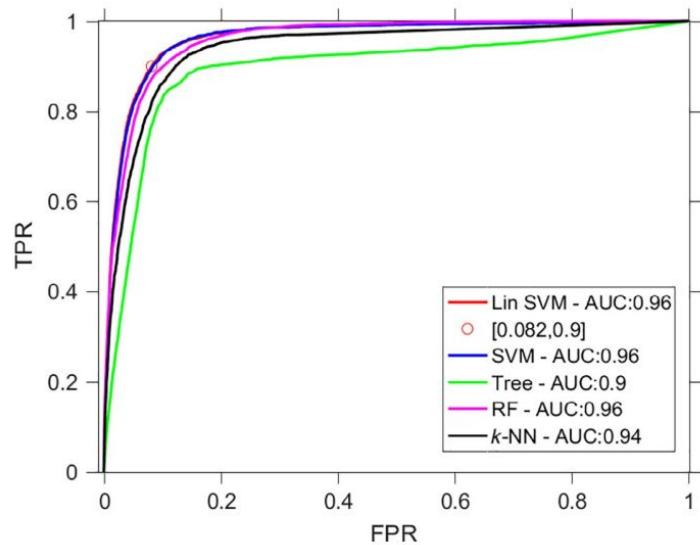


TABLE 2 Mean and Standard Deviation Computation Time (in Seconds) Necessary to Train a Classifier With the Cost-Sensitive Approach, and Mean Ratio With Respect to Linear SVM Computing Time

	Linear SVM	SVM	Tree	Random Forest	k-NN
Mean (in s)	106.8	1904.4	33.3	284.3	43.1
SD (in s)	32.3479	654.463	8.669	162.8966	9.7101
Ratio w.r.t. linear SVM	1	18.8013	0.3354	2.8544	0.4288

Notes: The mean and standard deviations are computed over 50 runs. Calculations are done on an Intel(R) Xeon(R) processor with 10 cores, a CPU at 2.80 GHz, and 256 GB RAM.

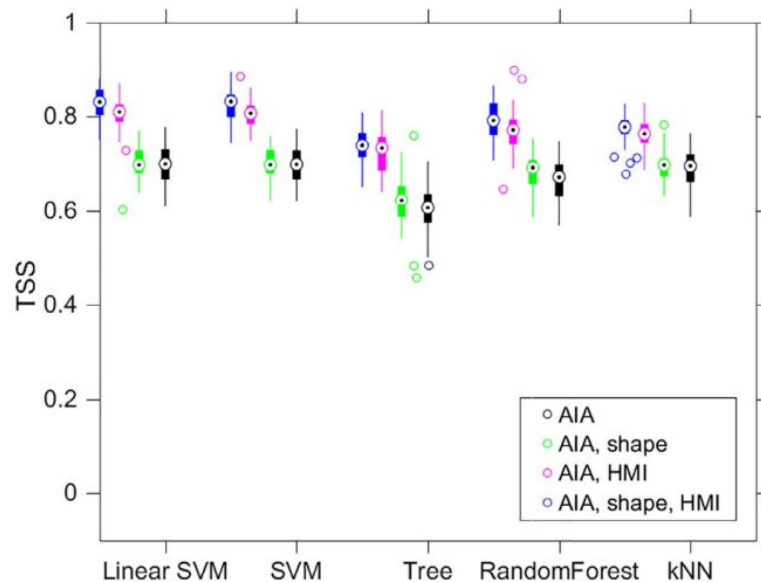


Performance: SVM>ADABoost>RUSBoost

Importance of Attributes

Magnetogram improves Performance.

Shape information is less critical.



Kasapis, S., Zhao, L., Chen, Y., Wang, X., Bobra, M., & Gombosi, T. (2022). **Interpretable machine learning to forecast SEP events for Solar Cycle 23.** *Space Weather*, 20, e2021SW002842.

Use machine learning methods to predict whether an active region (AR) which produces flares will lead to a solar energetic particle (SEP) event.

SEP events can cause space storms that may harm valuable assets in space, therefore it is important to predict their occurrence accurately.

Focused on using the SMARP data set to underline the contrast between the flares that lead to SEPs and flares that do not

Survey the space-Weather Michelson Doppler Imager (MDI) Active Region Patches (SMARPs) active regions associated with flares that appear on the solar disk between 5 June 1996 and 14 August 2010, label those that produced SEPs as positive and the rest as negative

Database

The SMARPs:

Tracked Active Region Patch (TARP) Number

Active region number,

Total unsigned flux (USFLUXL), mean gradient of the vertical field (MEANGBL), and the logarithm of the total unsigned flux near polarity inversion line (RVALUE)

Four spatial features

The SEP event list used in this work is documented in the NOAA Space Environment Service Center website.

Algorithms:

Four version of SVM: one linear kernel, two polynomial kernel, and one Gaussian Radial Basis Function kernel. Training data length is $n = 116$ and the maximum feature vector length is $p = 7$

Linear models: ridge regression (not adopting a cross-validation procedure for selecting the tuning parameter α due to considerations of samplesparsity)

Default Logistic Regression module provided by the Scikit-Learn library in Python

Training:

Training testing ratio set to be 9:1. Resample time $k=100$.

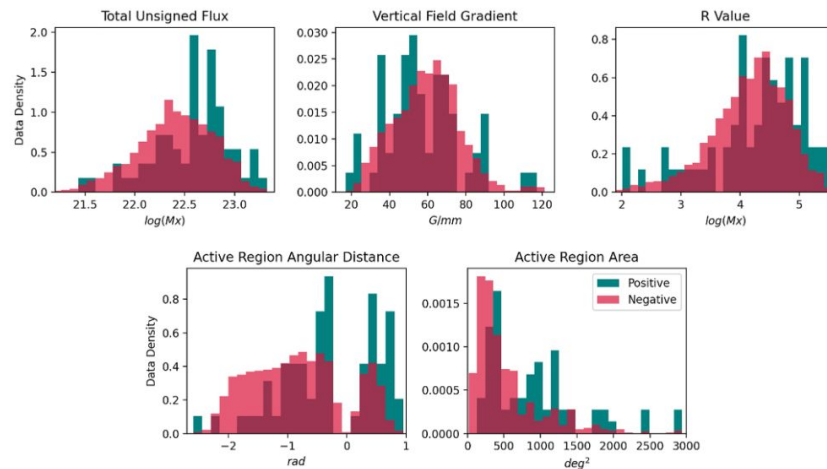
Three metrics are used to evaluate the
Performance of features and models:

Accuracy (ACC),

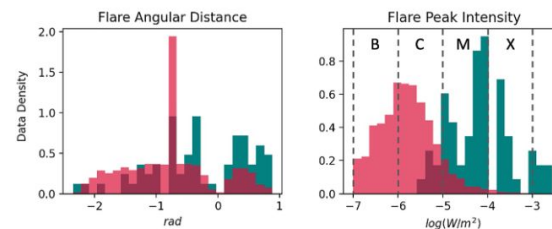
True Skill Statistics

Heidke Skill Score.

SMARP Predictors



Flare Predictors



The optimal decision threshold is found out to be 0.5

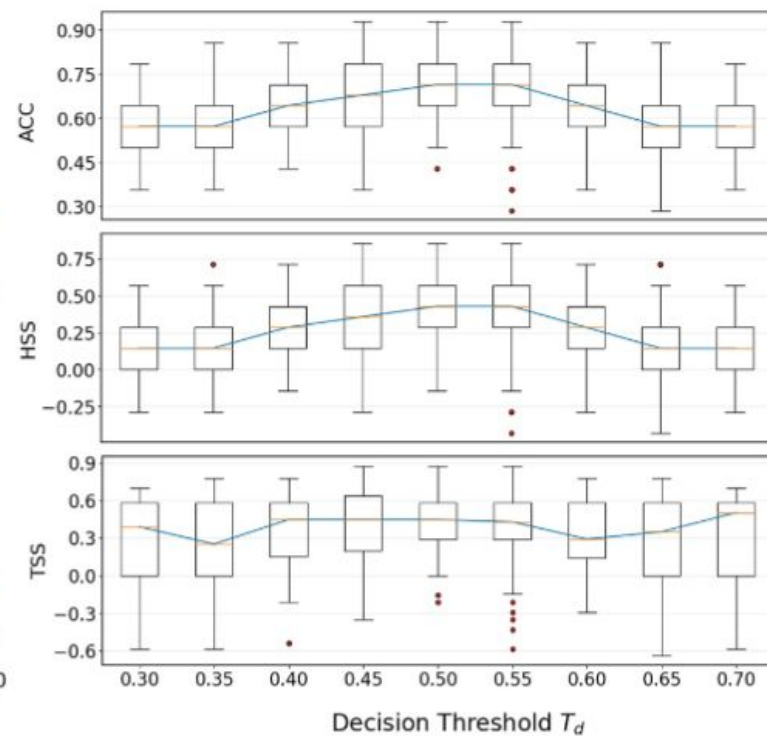
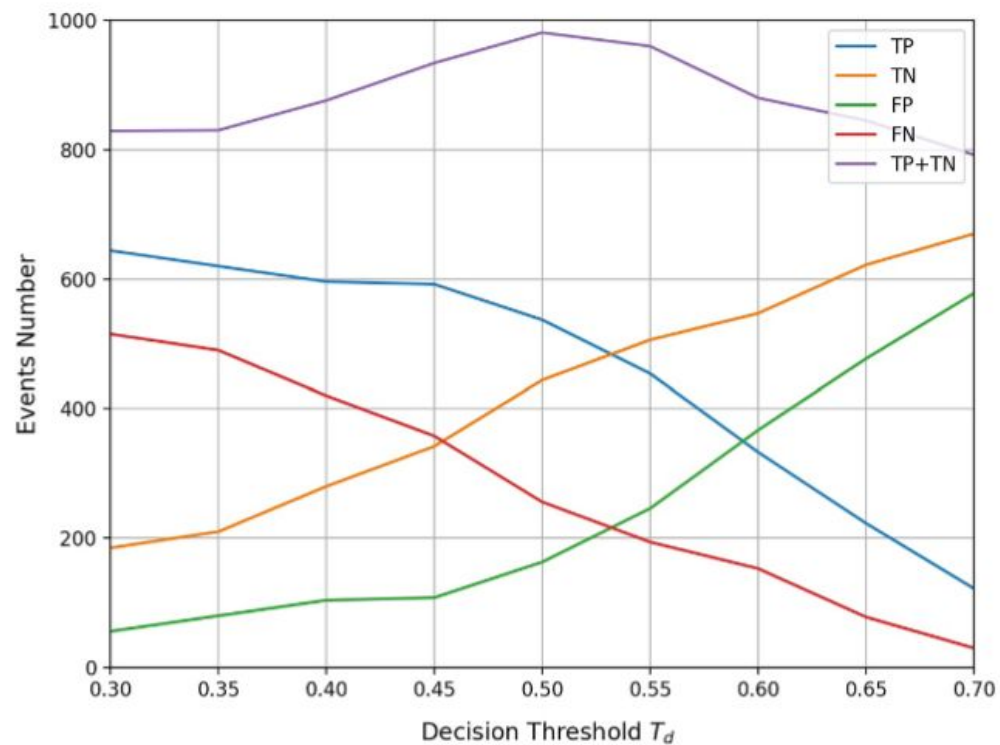


Table A1*Maximum Accuracy (ACC), True Skill Statistics (TSS) and Heidke Skill Score (HSS) Values for the SVM and Linear Models Using SMARP Predictors*

SMARP Predictors	SVMs				Linear Models		Score
	Linear	RBF	Polynomial 2	Polynomial 3	Logistic Reg.	Ridge	
1. USFLUXL & ARDIST	0.67 ± 0.12	0.67 ± 0.13	0.70 ± 0.12	0.72 ± 0.12	0.70 ± 0.12	0.71 ± 0.12	ACC
	0.39 ± 0.28	0.38 ± 0.28	0.43 ± 0.25	0.47 ± 0.24	0.43 ± 0.23	0.47 ± 0.25	TSS
	0.34 ± 0.24	0.34 ± 0.25	0.39 ± 0.23	0.44 ± 0.25	0.40 ± 0.24	0.42 ± 0.24	HSS
2. USFLUXL & ARAREA	0.65 ± 0.11	0.67 ± 0.12	0.65 ± 0.12	0.67 ± 0.12	0.69 ± 0.11	0.65 ± 0.12	ACC
	0.35 ± 0.27	0.38 ± 0.28	0.34 ± 0.27	0.36 ± 0.26	0.37 ± 0.23	0.30 ± 0.27	TSS
	0.30 ± 0.23	0.35 ± 0.24	0.30 ± 0.24	0.33 ± 0.24	0.34 ± 0.22	0.27 ± 0.23	HSS
3. ARDIST & ARAREA	0.69 ± 0.11	0.69 ± 0.11	0.52 ± 0.04	0.51 ± 0.03	0.67 ± 0.12	0.70 ± 0.12	ACC
	0.42 ± 0.25	0.42 ± 0.23	0.15 ± 0.28	0.10 ± 0.22	0.36 ± 0.25	0.42 ± 0.26	TSS
	0.37 ± 0.23	0.38 ± 0.22	0.05 ± 0.09	0.03 ± 0.06	0.34 ± 0.24	0.40 ± 0.25	HSS
4. ARDIST & RVALUE	0.65 ± 0.13	0.68 ± 0.10	0.58 ± 0.11	0.60 ± 0.11	0.67 ± 0.11	0.66 ± 0.12	ACC
	0.33 ± 0.28	0.38 ± 0.22	0.18 ± 0.29	0.26 ± 0.29	0.36 ± 0.23	0.35 ± 0.25	TSS
	0.31 ± 0.26	0.35 ± 0.20	0.15 ± 0.22	0.21 ± 0.23	0.34 ± 0.21	0.33 ± 0.23	HSS
5. USFLUXL, ARDIST & ARAREA	0.67 ± 0.13	0.68 ± 0.11	0.70 ± 0.11	0.67 ± 0.10	0.70 ± 0.13	0.69 ± 0.10	ACC
	0.36 ± 0.28	0.38 ± 0.24	0.42 ± 0.23	0.37 ± 0.21	0.42 ± 0.27	0.41 ± 0.22	TSS
	0.34 ± 0.26	0.35 ± 0.22	0.39 ± 0.22	0.34 ± 0.19	0.39 ± 0.26	0.38 ± 0.21	HSS
6. All SMARP Predictors	0.68 ± 0.12	0.68 ± 0.13	0.66 ± 0.13	0.69 ± 0.12	0.67 ± 0.11	0.69 ± 0.13	ACC
	0.40 ± 0.24	0.32 ± 0.27	0.35 ± 0.28	0.42 ± 0.25	0.36 ± 0.24	0.40 ± 0.27	TSS
	0.37 ± 0.23	0.35 ± 0.25	0.33 ± 0.27	0.39 ± 0.24	0.33 ± 0.23	0.38 ± 0.26	HSS

Note. The ACC values ≥ 0.70 are marked in bold. In green and red are marked the higher and lower accuracy values respectively.

Table A2
Maximum Accuracy (ACC), True Skill Statistics (TSS) and Heidke Skill Score (HSS) Values for the Support Vector Machines (SVM) and Linear Models Using Different Predictors

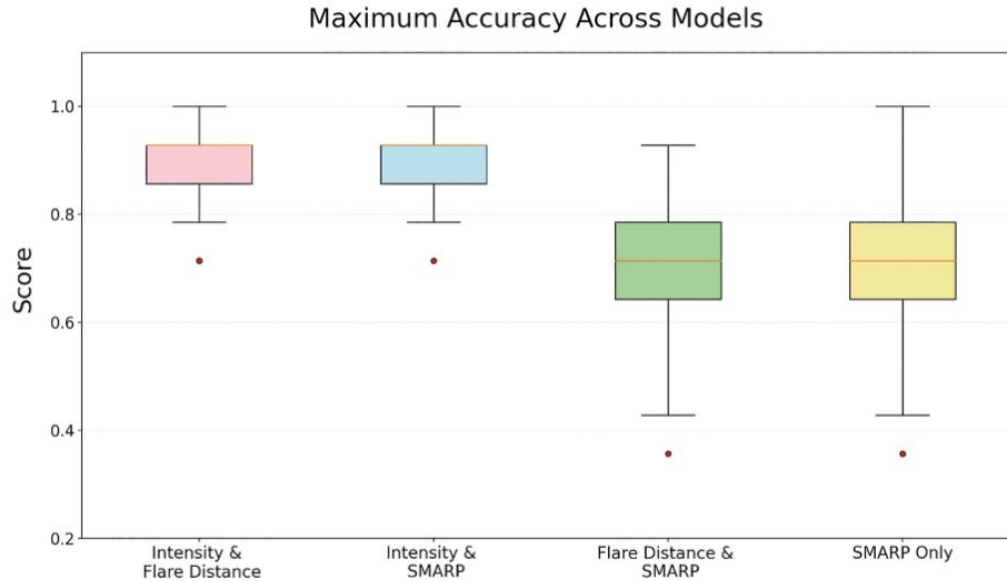
Flare Predictors	SVMs				Linear Models		
	Linear	RBF	Polynomial 2	Polynomial 3	Logistic Reg.	Ridge	
1. Intensity & Flare Distance	0.90 ± 0.08	0.91 ± 0.07	0.90 ± 0.08	0.90 ± 0.08	0.90 ± 0.08	0.91 ± 0.07	ACC
	0.82 ± 0.16	0.84 ± 0.13	0.82 ± 0.15	0.80 ± 0.16	0.83 ± 0.16	0.84 ± 0.12	TSS
	0.80 ± 0.17	0.82 ± 0.14	0.80 ± 0.16	0.78 ± 0.17	0.80 ± 0.17	0.82 ± 0.14	HSS
SMARP & Flare Predictors							
2. Flare Distance & USFLUXL	0.71 ± 0.10	0.68 ± 0.13	0.70 ± 0.11	0.70 ± 0.11	0.71 ± 0.11	0.71 ± 0.12	ACC
	0.46 ± 0.20	0.39 ± 0.28	0.44 ± 0.24	0.42 ± 0.24	0.45 ± 0.22	0.46 ± 0.25	TSS
	0.42 ± 0.20	0.36 ± 0.26	0.40 ± 0.23	0.39 ± 0.23	0.42 ± 0.21	0.41 ± 0.23	HSS
3. Flare Distance & RVALUE	0.67 ± 0.14	0.69 ± 0.13	0.61 ± 0.14	0.61 ± 0.12	0.69 ± 0.12	0.70 ± 0.10	ACC
	0.36 ± 0.29	0.41 ± 0.27	0.25 ± 0.32	0.30 ± 0.32	0.41 ± 0.25	0.43 ± 0.22	TSS
	0.34 ± 0.27	0.38 ± 0.26	0.22 ± 0.27	0.23 ± 0.23	0.38 ± 0.24	0.40 ± 0.21	HSS

Table A2
Continued

Flare Predictors	SVMs				Linear Models		
	Linear	RBF	Polynomial 2	Polynomial 3	Logistic Reg.	Ridge	
4. Flare Distance & ARAREA	0.66 ± 0.13	0.68 ± 0.11	0.60 ± 0.09	0.62 ± 0.10	0.70 ± 0.12	0.69 ± 0.11	ACC
	0.35 ± 0.27	0.40 ± 0.25	0.36 ± 0.30	0.39 ± 0.29	0.43 ± 0.24	0.42 ± 0.24	TSS
	0.32 ± 0.26	0.36 ± 0.22	0.19 ± 0.18	0.24 ± 0.21	0.41 ± 0.23	0.39 ± 0.22	HSS
5. Flare Distance, USFLUXL & ARAREA	0.69 ± 0.13	0.69 ± 0.13	0.67 ± 0.12	0.66 ± 0.11	0.69 ± 0.13	0.69 ± 0.11	ACC
	0.41 ± 0.27	0.42 ± 0.27	0.35 ± 0.30	0.36 ± 0.25	0.35 ± 0.26	0.41 ± 0.24	TSS
	0.38 ± 0.26	0.37 ± 0.26	0.32 ± 0.28	0.32 ± 0.23	0.39 ± 0.25	0.38 ± 0.22	HSS
6. Intensity & USFLUXL	0.88 ± 0.09	0.89 ± 0.07	0.90 ± 0.08	0.90 ± 0.08	0.88 ± 0.09	0.89 ± 0.09	ACC
	0.80 ± 0.16	0.80 ± 0.13	0.82 ± 0.14	0.80 ± 0.17	0.78 ± 0.17	0.79 ± 0.17	TSS
	0.77 ± 0.18	0.77 ± 0.15	0.80 ± 0.15	0.79 ± 0.18	0.76 ± 0.18	0.77 ± 0.17	HSS
7. Intensity & RVALUE	0.89 ± 0.08	0.91 ± 0.07	0.91 ± 0.08	0.89 ± 0.09	0.91 ± 0.07	0.90 ± 0.08	ACC
	0.79 ± 0.16	0.83 ± 0.14	0.83 ± 0.15	0.80 ± 0.16	0.84 ± 0.13	0.83 ± 0.15	TSS
	0.77 ± 0.17	0.82 ± 0.15	0.81 ± 0.16	0.78 ± 0.17	0.81 ± 0.14	0.80 ± 0.16	HSS
8. Intensity & ARDIST	0.91 ± 0.07	0.91 ± 0.07	0.91 ± 0.07	0.90 ± 0.07	0.92 ± 0.07	0.91 ± 0.07	ACC
	0.84 ± 0.13	0.84 ± 0.13	0.83 ± 0.14	0.83 ± 0.13	0.86 ± 0.13	0.84 ± 0.14	TSS
	0.82 ± 0.14	0.81 ± 0.14	0.81 ± 0.15	0.81 ± 0.14	0.84 ± 0.15	0.82 ± 0.15	HSS
9. Intensity, USFLUXL & ARDIST	0.91 ± 0.07	0.91 ± 0.08	0.90 ± 0.08	0.91 ± 0.07	0.92 ± 0.08	0.90 ± 0.08	ACC
	0.83 ± 0.14	0.84 ± 0.15	0.82 ± 0.15	0.83 ± 0.13	0.85 ± 0.15	0.81 ± 0.16	TSS
	0.82 ± 0.15	0.82 ± 0.16	0.80 ± 0.16	0.82 ± 0.14	0.83 ± 0.16	0.80 ± 0.17	HSS
10. Intensity & MEANGBL	0.90 ± 0.08	0.91 ± 0.07	0.91 ± 0.08	0.90 ± 0.08	0.91 ± 0.08	0.90 ± 0.08	ACC
	0.82 ± 0.15	0.85 ± 0.13	0.84 ± 0.15	0.82 ± 0.15	0.84 ± 0.14	0.82 ± 0.16	TSS
	0.80 ± 0.17	0.83 ± 0.14	0.83 ± 0.15	0.80 ± 0.16	0.82 ± 0.15	0.80 ± 0.16	HSS

Note. The ACC values ≥ 0.91 with standard deviation ≤ 0.07 are marked in bold. In green and red are marked the higher and lower accuracy values respectively for each one of the three predictor groups.

Results



Prediction results for the SMARP data set (71%) are worse than the ones produced using the are peak intensity and location(91%), but it provides a better leading time.